THE LANGUAGE FAMILY EFFECT: IMPROVING AFRICAN SENTIMENT MODELS THROUGH LINGUISTIC RELATEDNESS

Abstract

African languages represent one of the world's most linguistically diverse regions, yet they remain critically under-resourced in Natural Language Processing, limiting the development of equitable and effective language technologies. Sentiment analysis for these languages is particularly constrained by scarce labeled data, limited representation in pretrained models, and heavy reliance on translation-based pipelines that introduce cultural and semantic distortion, especially in code-switched contexts. This work extends the AfriSenti benchmark with sentiment data from 38 additional African languages and examines how linguistic relatedness, captured through language family structure, can be leveraged to improve multilingual sentiment modeling. We evaluate two complementary approaches: extended task-adaptive pretraining on a large and heterogeneous multilingual dataset, and direct language and family conditioning through explicit input-level metadata. Experiments conducted at the overall, language family, and individual language levels show that task-adaptive pretraining provides strong gains in large and noisy multilingual settings, while direct language and family conditioning is most effective on smaller and cleaner benchmarks such as AfriSenti. Together, these results provide empirical evidence for the language family effect, demonstrating that both implicit and explicit modeling of genealogical relationships improves robustness and generalization for African languages that are underrepresented or absent in standard pretrained models. Our findings highlight the value of linguistically grounded and data-efficient approaches for building more inclusive and sustainable NLP systems for African languages.

Keywords: African Languages, Sentiment Analysis, Language Families, Multilingual NLP

INTRODUCTION

Africa is home to more than 2,000 languages spoken by over 1.3 billion people, making it one of the most linguistically diverse regions in the world (Cervone, 2025). Despite this diversity, Natural Language Processing research for African languages remains significantly underdeveloped when compared to high-resource languages such as English, Chinese, and Spanish. This imbalance is driven by a combination of historical marginalization, limited digital infrastructure, and insufficient availability of large, high-quality text corpora. As a result, many African languages are either poorly represented or entirely absent from widely used NLP benchmarks and pretrained language models, which limits the applicability and reliability of current NLP technologies in African contexts.

Sentiment analysis is a core NLP task with important applications in governance, public health, e-commerce, and social media, but its effectiveness depends heavily on the availability of labeled training data, which remains scarce for most African languages. The AfriSenti SemEval Shared Task (2023) marked a major advance by releasing annotated sentiment datasets for 14 African languages and establishing a benchmark for systematic evaluation. Subsequent work shows that African language–specific transformer models often outperform general multilingual models in low-resource settings, while also revealing the limitations of translation-based zero-shot approaches and the continued relevance of lightweight methods. Building on these insights, this paper presents a comprehensive study of sentiment analysis across a broader set of African languages, with a focus on linguistic

relatedness and language family structure. We extend the AfriSenti benchmark with sentiment data from 38 additional African languages and incorporate code-switched text to better reflect real-world communication. To examine the language family effect, we evaluate two complementary strategies: extended task-adaptive pretraining, which leverages large-scale multilingual sentiment data to improve cross-lingual representations, and direct language–family conditioning, which explicitly encodes language identity and genealogical information at the input level. Model performance is analyzed at the overall, language family, and individual language levels, both before and after fine-tuning on AfriSenti, allowing us to isolate the contributions of each approach and demonstrate how language family awareness improves robustness and generalization in low-resource African sentiment analysis.

RELATED STUDIES

Pretrained Models for African Languages

Natural Language Processing research for African languages has historically lagged behind work on high-resource languages due to limited data availability, uneven digital representation, and restricted computational resources. Early approaches often relied on generic multilingual models trained predominantly on high-resource languages, resulting in degraded and unstable performance when applied to African languages. As the field has progressed, studies have increasingly highlighted the limitations of general-purpose multilingual architectures such as multilingual BERT and XLM-R, which struggle to capture the morphological complexity, lexical diversity, and orthographic variation characteristic of many African languages. These limitations have motivated the development of transformer-based models pretrained directly on African language corpora to improve representation learning through linguistically relevant data exposure.

AfriBERTa, introduced by Ogueji (2021), represents a foundational effort in African language specific pretraining and demonstrates clear improvements over multilingual BERT on downstream tasks including named entity recognition and sentiment analysis seen here Aryal (2023). Subsequent models such as AfroXLM-R and AfroLM (Dossou, 2022) further reinforce these findings, showing that in-domain pretraining yields more robust and transferable representations across African languages. Despite these advances, full-scale pretraining remains computationally expensive and inaccessible for many researchers, while the scarcity of labeled datasets and the prevalence of code-switching continue to pose significant challenges. Building on this prior work, our approach combines efficient model adaptation, expanded sentiment resources, and family-aware training strategies to address these limitations and advance sentiment analysis across a broader range of African languages.

Cross-Lingual Transfer and Language Typology

A growing body of research has examined the role of cross-lingual transfer and language typology in improving NLP performance for low-resource languages. The central premise of this work is that languages belonging to the same genealogical family share common linguistic properties, including vocabulary roots, morphological patterns, syntactic structures, and semantic regularities. Models trained on one or more source languages can exploit these shared characteristics to improve generalization when applied to related target languages, particularly in settings where labeled data for the target language is scarce or unavailable. This family-aware perspective has become especially relevant for multilingual regions such as Africa, where many languages are closely related yet unevenly represented in NLP

datasets. For example, prior work has demonstrated that leveraging linguistic similarity between closely related languages such as Kirundi and Kinyarwanda can significantly improve performance at the single-language level, providing early empirical evidence for the effectiveness of family-based transfer in African NLP (Marivate, 2024).

Empirical evidence strongly supports the effectiveness of typology-aware transfer strategies across a range of NLP tasks. According to Kim (2017), cross-lingual transfer for Part-of-Speech tagging can be achieved without parallel corpora or machine translation, and selecting source languages from the same language family as the target language leads to significant performance gains. Similarly, Taghizadeh (2022) investigated cross-lingual transfer for relation extraction using Universal Dependencies and found that linguistic similarity between source and target languages is a strong predictor of successful transfer. Expanding on this line of work, Lin (2019) conducted a comprehensive analysis of transfer language selection and showed that while phylogenetic distance and language family membership are important factors, other considerations such as lexical overlap, syntactic similarity, domain alignment, and data quality also influence transfer effectiveness. Together, these studies highlight the importance of incorporating linguistic relatedness into model design and training strategies, particularly for multilingual and low-resource NLP settings.

METHODOLOGY

Dataset Expansion and Preprocessing

To increase the linguistic coverage and contextual diversity of our training data, we expanded the original AfriSenti corpus (Shamsuddeen Hassan Muhammad I. A., 2023) with additional sentiment-labeled datasets drawn from a wide range of African languages. Building on this effort, we incorporated sentiment-labeled datasets from Owusu (2025), selecting 38 African languages to expand the linguistic and genealogical diversity of the AfriSenti corpus. These languages were chosen to improve coverage across underrepresented language families and include data drawn from multiple sources, such as social media platforms, online news articles, and weakly supervised data collection efforts. As a result of this expansion, the training set grew substantially from 63,685 to 431,021 samples.

All collected datasets were subjected to a unified preprocessing pipeline to ensure consistency across sources and languages. Sentiment labels originating from different annotation schemes were mapped to a unified three-class label set consisting of positive, negative, and neutral. For languages with very large datasets, where the expanded data predominantly followed a binary sentiment labeling scheme, we applied balanced sampling by selecting an equal number of positive and negative examples and capped the number of training instances at 10,000 per language. We also added 5,000 samples per language to the full test set alongside the AfriSenti Dataset. This preprocessing strategy ensured balanced, high-quality data suitable for systematic cross-lingual and family-aware analysis.

af	**am**	**ak**	**kab**	**sg**	**ts**	**tmh**	**ff**
bm	**ny**	**ig**	**kg**	**umb**	**tn**	**ha**	**om**
bem	**fon**	**nso**	**zu**	**sw**	**wo**	**ss**	**yo**

lg	**mos**	**tum**	**kea**	**so**	**ln**	**dyu**	**kam**
xh	**tw**	**kmb**	**ee**	**kbp**	**rn**		

Table 1: List of the 38 African languages included in the expanded sentiment analysis dataset, represented by ISO 639 language codes.

Language Family Analysis

To examine the impact of linguistic typology on model performance, we mapped each language in the combined dataset to its corresponding language family using established linguistic classifications. This grouping allows us to analyze whether genealogical relatedness influences cross-lingual transfer and sentiment classification performance. The 38 languages in our dataset span four major language families commonly represented in Africa. In the following grouping, language names refer to those included in the original AfriSenti dataset, while ISO language codes correspond to additional languages introduced through our expanded dataset.

- Afro-Asiatic (Amharic, Algerian Arabic, Moroccan Arabic, Hausa, Oromo, Tigrinya): so, om, ha, am, kab, tmh
- Niger-Congo (Igbo, Kinyarwanda, Swahili, Tsonga, Twi, Yoruba): xh, tn, ny, lg, ln, nso, rn, ee, wo, tum, zu, bem, kg, mos, ss, umb, ff, kmb, bm, dyu, fon, kam, yo, tw, sw, ts, ig, kbp, ak
- Creole (Nigerian Pidgin): sg, kea
- Indo-European (Mozambican Portuguese): af

This family-aware categorization provides a structured foundation for our multilingual training and evaluation experiments. It enables systematic investigation of cross-lingual transfer effects both within individual language families and across typologically distinct families, supporting a deeper analysis of how linguistic relatedness contributes to model robustness and generalization.

Modeling

All experiments use mmBERT-base, a state-of-the-art BERT model based on ModernBERT and trained in a highly multilingual context that includes several African languages (Durme, 2025), making it well suited for evaluating multilingual and cross-lingual sentiment analysis in low-resource settings.

Extended Task Adaptive Pretraining (ETAP)

The modeling pipeline consists of two stages: an initial training phase on the expanded multilingual sentiment dataset to learn broad cross-lingual representations, followed by targeted fine-tuning on the AfriSenti dataset to adapt the model to the benchmark's specific task and data distribution.

Task-adaptive pretraining was performed on the combined multilingual sentiment dataset to expose the model to a diverse range of African languages and sentiment expressions. All corpora were shuffled to ensure a balanced distribution of languages and labels during

optimization. We used a maximum sequence length of 64 tokens, reflecting the short-form nature of social media and conversational text while enabling larger batch sizes and more stable training. An effective batch size of 2,304 was achieved through gradient accumulation, following empirical evaluation of multiple batch configurations, and was found to yield the best performance. We hypothesize that the expanded and heterogeneous dataset benefits from larger batch sizes, which reduce gradient variance and improve generalization across low-resource languages. Training was conducted for 10 epochs using the AdamW optimizer with a learning rate of 5e-5 and linear warmup over 6 percent of training steps. Weight decay and label smoothing were applied to improve generalization, and early stopping with a patience of two epochs was used to prevent overfitting.

In the second stage, the model was directly fine-tuned on the AfriSenti dataset to adapt the representations learned during task-adaptive pretraining to the specific characteristics of the benchmark. We retained a maximum sequence length of 64 tokens to maintain consistency with the pretraining stage and to accommodate the short, informal nature of sentiment-bearing text. Fine-tuning was performed with a batch size of 256, which balances optimization stability with memory efficiency, and a learning rate of 1e-4 to enable rapid adaptation while avoiding catastrophic forgetting. Weight decay was set to 0.01 and label smoothing to 0.05 to promote generalization and reduce overconfidence on limited labeled data. A linear warmup schedule was applied to 5 percent of training steps to stabilize early optimization. Models were trained for up to 10 epochs with early stopping enabled and a patience of one epoch, reflecting the relatively small size of the AfriSenti dataset and the need to prevent overfitting during task-specific adaptation.

Direct Language and Family Conditioning (DLFC)

We conducted a set of experiments in which direct language and family information was incorporated during fine-tuning on the AfriSenti dataset. In this setting, we augmented the tokenizer vocabulary with three special tokens, [LANG], [FAMILY], and [TEXT], which were prepended to each input sequence to explicitly encode the language identity and language family of the text. This input-level conditioning allows the model to directly access linguistic metadata when making sentiment predictions, rather than relying solely on implicit distinctions learned through shared token embeddings. For example, an input sequence was formatted as “[LANG] Portuguese [FAMILY] Indo-European [TEXT] Esse produto é maravilhoso”. All training hyperparameters and optimization settings were kept identical to the baseline fine-tuning configuration to ensure a controlled comparison and isolate the effect of direct linguistic conditioning.

Combined Approach

In addition to applying direct language and family conditioning during AfriSenti fine-tuning, we explored a combined approach in which the same input augmentation strategy was applied throughout the full two-stage training pipeline. In this setting, language and family tokens were added not only during AfriSenti fine-tuning but also during the initial task-adaptive pretraining phase on the expanded multilingual dataset. This approach enables the model to jointly learn sentiment representations and linguistic structure from a larger and more diverse corpus while being consistently conditioned on language identity and genealogical relationships. By integrating direct linguistic metadata across both training

stages, this combined approach provides a more comprehensive evaluation of the language family effect and its influence on model robustness and cross-lingual generalization.

Evaluation

Model evaluation is conducted on the expanded multilingual sentiment dataset, with performance additionally assessed on the AfriSenti benchmark to enable direct comparison with prior work. To support robust and fair evaluation, we augmented the test set by adding up to 5,000 held-out samples per language from the expanded dataset, which were evaluated alongside the AfriSenti test data. Evaluation on the expanded dataset is performed prior to direct fine-tuning on AfriSenti, allowing us to measure the quality of representations learned during task-adaptive pretraining without influence from benchmark-specific adaptation.

Performance is evaluated using standard classification metrics, with weighted F1-score reported as the primary measure to account for class imbalance and capture the trade-off between precision and recall. Results on the expanded dataset are analyzed in terms of overall performance across all languages, performance aggregated by language family, and top-performing individual languages. We evaluate overall performance, language level and language family performance. Language-level performance is reported for the top five and bottom five languages, selected based on average F1-score across models, in order to highlight both strong and weak cases and provide insight into how different training strategies affect performance across diverse linguistic and resource conditions.

Following this initial evaluation, models are directly fine-tuned on the AfriSenti dataset and re-evaluated under the same metric framework. Performance on AfriSenti is compared against a baseline obtained by directly fine-tuning mmBERT on AfriSenti without task-adaptive pretraining or direct language and family conditioning. This staged evaluation design allows us to isolate the contributions of expanded multilingual training and family-aware modeling while maintaining a controlled comparison on a standardized benchmark.

RESULTS & DISCUSSION

Overall Performance

Model Configuration	Class	Precision	Recall	F1-Score
DLFC	Negative	0.4825	**0.8116**	**0.6052**
	Neutral	0.1730	0.0028	0.0055
	Positive	0.5252	0.2367	0.3263
	Wtd Avg	0.4841	0.4913	0.4374
ETAP	Negative	**0.5192**	0.5895	0.5521
	Neutral	**0.5574**	**0.3755**	0.4487
	Positive	0.5351	**0.4856**	**0.5091**

Model Configuration	Class	Precision	Recall	F1-Score
	Wtd Avg	**0.5290**	**0.5275**	**0.5256**

Table 2: Overall model performance on Expanded Test Set using both DLFC (language-family conditioning) and ETAP (task-adaptive pretraining). Weighted Average (Wtd Avg)

Table 2 presents overall sentiment classification performance on the expanded test set for models trained using direct language-family conditioning (DLFC) and extended task-adaptive pretraining (ETAP). The results reveal a substantial performance gap between the two approaches. ETAP consistently outperforms DLFC across all sentiment classes, achieving a weighted F1-score of 0.5256 compared to 0.4374 for DLFC. Notably, ETAP demonstrates balanced performance across negative, neutral, and positive classes, with particularly strong gains in neutral sentiment classification, where DLFC exhibits severe degradation. While DLFC attains high recall for the negative class, this comes at the cost of extremely low neutral recall, indicating that the model collapses toward predicting dominant sentiment classes under increased linguistic diversity. These findings suggest that explicit language and family metadata alone is insufficient to handle the complexity introduced by large-scale multilingual expansion.

In contrast, ETAP benefits from continued exposure to diverse multilingual data during task-adaptive pretraining, enabling the model to learn more stable and transferable sentiment representations without relying on explicit linguistic conditioning. The absence of additional metadata tokens in ETAP may reduce input complexity and allow the model to focus on sentiment-relevant patterns that generalize across languages. We hypothesize that this contributes to its improved robustness and higher overall performance on the expanded test set. Together, these results indicate that task-adaptive pretraining is more effective than language-family conditioning alone when scaling sentiment models to a large and heterogeneous set of African languages, and that explicit linguistic signals may require further calibration or architectural support to be beneficial in highly multilingual settings.

Performance by Language Family

Language Family	Instance # (N)	DLFC (F1)	ETAP (F1)	Perf. Diff
Indo-European	8,628	0.4373	**0.7506**	+0.3133
Creole	13,234	0.5180	**0.5477**	+0.0297
Afro-Asiatic	35,122	0.4403	**0.5107**	+0.0704
Niger-Congo	134,434	0.4205	**0.5105**	+0.0900

Table 3: Language Family level performance on Expanded Test Set using both DLFC (language-family conditioning) and ETAP (task-adaptive pretraining).

Table 3 presents sentiment classification performance aggregated at the language family level on the expanded test set, comparing language-family conditioning (DLFC) with task-adaptive pretraining (ETAP). Across all four language families, ETAP consistently outperforms DLFC, demonstrating its stronger ability to generalize across linguistically diverse groups.

The largest performance gain is observed for Indo-European languages, where ETAP achieves an F1-score of 0.7506 compared to 0.4373 for DLFC. This substantial improvement suggests that task-adaptive pretraining is particularly effective at leveraging shared sentiment patterns across languages that may be structurally distinct from the majority of African languages present in the training data. In contrast, DLFC appears less capable of transferring effectively at the family level, likely due to its reliance on explicit metadata rather than robust shared representations.

More moderate but consistent gains are observed for Creole, Afro-Asiatic, and Niger-Congo language families. While the performance difference for Creole languages is relatively small, this may reflect the limited number of languages and more homogeneous structure within this family. For Afro-Asiatic and Niger-Congo, which together account for the majority of languages and instances in the dataset, ETAP demonstrates clear improvements despite substantial linguistic variation and data imbalance. Importantly, these gains are achieved even though many individual languages within these families are not present in the original AfriSenti dataset. This indicates that ETAP facilitates effective cross-family and cross-language transfer by learning generalized sentiment representations during expanded multilingual training. Overall, these results support the conclusion that task-adaptive pretraining provides a more scalable and robust approach to multilingual sentiment modeling than language-family conditioning alone, particularly when extending to previously unseen languages and families.

Language Level Performance

Language	Instance # (N)	DLFC (F1)	ETAP (F1)	Perf. Diff
Afrikaans	4,966	0.6891	**0.7821**	+0.0930
Nigerian Pidgin	4,154	0.6204	**0.6446**	+0.0242
Somali	4,969	0.5743	**0.6545**	+0.0802
Algerian Arabic	958	0.5432	**0.6485**	+0.1053
Swahili	5,711	0.4976	**0.6873**	+0.1897
Igbo	8,412	0.3523	**0.4539**	+0.1016
Oromo	7,029	0.3901	**0.4213**	+0.0312
Yoruba	9,225	0.3103	**0.5077**	+0.1974
Hausa	10,227	0.3981	**0.4420**	+0.0439
Portuguese	3,662	0.1565	**0.7099**	+0.5534

Table 4: Language Level performance on Expanded Test Set using both DLFC (language-family conditioning) and ETAP (task-adaptive pretraining). Reporting the Top 5 and Bottom 5 languages.

Table 4 presents language-level sentiment classification results on the expanded test set, reporting the top and bottom five languages based on average F1-score across models. Across all languages, task-adaptive pretraining (ETAP) consistently outperforms direct language–family conditioning (DLFC), demonstrating positive performance gains in every case. Among the top-performing languages, ETAP yields substantial improvements for Swahili and Algerian Arabic, indicating stronger generalization even across typologically and regionally diverse languages, while maintaining consistent gains for languages such as Nigerian Pidgin. These improvements are even more pronounced among the lowest-performing languages, where DLFC struggles despite large training instance counts, as seen in Yoruba and Igbo. The most striking gain occurs for Portuguese, where ETAP dramatically outperforms DLFC despite moderate data availability, highlighting ETAP's ability to mitigate representational gaps that explicit linguistic metadata alone cannot address. Overall, these results demonstrate that ETAP produces more stable and transferable sentiment representations across languages with varying resource levels, reinforcing the conclusion that task-adaptive pretraining is better suited than language-family conditioning alone for large-scale, heterogeneous multilingual sentiment analysis.

Performance on AfriSenti

Model Configuration	Precision	Recall	F1-Score	Perf. Gain
DLFC	**0.6352**	**0.6308**	**0.6314**	**+0.0250**
ETAP	0.6239	0.6232	0.6229	+0.0165
ETAP + DLFC	0.6251	0.6235	0.6239	+0.0175
Baseline	0.6072	0.6068	0.6064	-

Table 5: Overall Performance on AfriSenti Test Set using both DLFC (language-family conditioning), ETAP (task-adaptive pretraining), and ETAP + DLFC.

Table 5 presents overall sentiment classification performance on the AfriSenti test set, comparing language-family conditioning (DLFC), task-adaptive pretraining (ETAP), their combined variant, and a baseline model obtained through direct fine-tuning. Among all configurations, DLFC achieves the highest performance across precision, recall, and F1-score, yielding an improvement of +0.0250 over the baseline. Both ETAP and the combined ETAP + DLFC approach also outperform the baseline, though to a lesser extent. These results indicate that all proposed modeling strategies contribute positively to performance on AfriSenti, with explicit language and family conditioning providing the strongest gains in this setting.

We hypothesize that the effectiveness of DLFC on AfriSenti stems from the relatively small and less noisy nature of the benchmark dataset. In this context, explicitly encoding language and family information allows the model to more effectively learn structured linguistic distinctions and leverage typological relationships without being overwhelmed by heterogeneous training signals. In contrast, while ETAP demonstrates strong performance on large and noisy multilingual datasets, its advantages are less pronounced when applied to a smaller, more controlled benchmark such as AfriSenti. The combined ETAP + DLFC

approach improves over the baseline but does not surpass DLFC alone, suggesting that the complexity introduced by extended multilingual training may limit the benefits of explicit conditioning in this setting. Overall, these findings highlight a trade-off between robustness to large-scale multilingual noise and the effectiveness of explicit linguistic signals, underscoring the importance of selecting modeling strategies based on dataset size and complexity.

Figure 1: *AfriSenti PR CURVE*

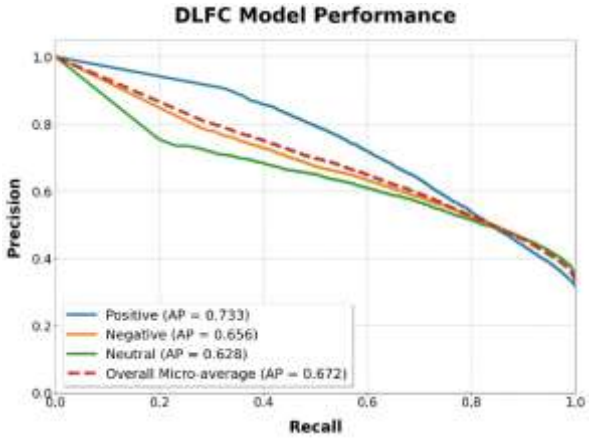

Figure 2: *DLFC PR CURVE on AfriSenti*

Figures 1 and 2 provide a more detailed view of model behavior on the AfriSenti test set and complement the aggregate results reported in Table 5. Figure 1 shows that DLFC consistently maintains higher precision across most recall thresholds compared to ETAP, ETAP + DLFC, and the baseline model, particularly in the low- to mid-recall range, indicating more reliable predictions when precision is prioritized. Although the differences in micro-averaged average precision are relatively small, the DLFC curve dominates for a substantial portion of the recall spectrum, reinforcing its overall performance advantage. Figure 2 further explains this trend by highlighting strong class-wise performance for DLFC, especially for positive and negative sentiment, where average precision is highest, while the neutral class remains the most challenging due to greater ambiguity. Together, these precision–recall curves demonstrate that DLFC's improvements on AfriSenti are consistent across decision thresholds and not limited to a single operating point, supporting the conclusion that DLFC is particularly effective in smaller, less noisy benchmark settings.

Discussion

The results of this study highlight that DLFC and ETAP provide complementary strengths for improving sentiment analysis across African languages, reflecting different ways in which linguistic information can enhance model performance. DLFC proves most effective in smaller, cleaner datasets such as AfriSenti, where explicit language and family information can be leveraged without being overwhelmed by noise. In these settings, directly conditioning the model on language and genealogical metadata introduces a strong inductive bias, allowing the model to more accurately capture sentiment patterns that vary across language families. ETAP, by contrast, demonstrates superior performance on large and heterogeneous datasets, where continued exposure to diverse multilingual data enables the model to learn robust and transferable sentiment representations across languages.

A central takeaway from these findings is that language family information consistently boosts performance for African languages, whether it is incorporated explicitly or learned implicitly. Many African languages are underrepresented or entirely absent in standard multilingual pretraining corpora, making it difficult for models such as mmBERT to generalize effectively based solely on shared embeddings. ETAP mitigates this limitation by strengthening internal representations through task-adaptive pretraining, allowing the model to infer latent similarities across languages and families. DLFC addresses the same challenge from a different perspective by explicitly signaling linguistic structure to the model, which is particularly advantageous when labeled data is limited and the dataset is well controlled.

These results also underscore important trade-offs associated with training a single shared model across many languages. DLFC reduces uncertainty by explicitly informing the model of the linguistic context it is processing, rather than requiring it to infer language identity and relationships solely from learned embeddings. This clarity improves stability and performance in low-noise settings but can become less effective as data diversity and noise increase. ETAP, while less interpretable, scales more effectively in such scenarios by learning language-agnostic sentiment representations that generalize across families. Together, these findings suggest that language family awareness is a critical component of African NLP, and that the choice between explicit conditioning and adaptive pretraining should be guided by dataset scale, noise, and deployment context rather than treated as a one-size-fits-all solution.

CONCLUSION

This work demonstrates that linguistic relatedness, particularly language family information, can be effectively leveraged to improve sentiment analysis for underrepresented African languages. By extending AfriSenti with data from 38 additional languages and evaluating task-adaptive pretraining and direct language family conditioning, we show that both approaches outperform standard fine-tuning. Task-adaptive pretraining is most effective in large and noisy multilingual settings, while direct language family conditioning yields stronger gains on smaller and cleaner benchmarks such as AfriSenti. These findings provide clear evidence for the language family effect and establish a practical framework for incorporating linguistic structure into multilingual sentiment modeling, supporting the development of more inclusive and effective NLP systems for African languages.

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